



Work in Progress: Dynamic Trust and Authority Assignment in Autonomous Multiagent Teams

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In this paper, we consider progress toward a systematic approach to managing a critical aspect of complex system uncertainty – the ability of participating agents, human or machine, to solve decision-making problems satisfactorily and in time. We propose a computable and actionable representation of system complexity as a collection of values that comprise solution tractability on a time budget and context-dependent measures of solution quality. The working hypothesis is that it is feasible to model agent’s decision-making capabilities as a function of context-dependent problem parameters and to use these models to predict and forestall transitions from safe and efficient operations to operations with degrading efficiency and safety. We outline the approach[†] and describe initial feasibility results using simulated decision makers. Initial results indicate that dynamic, computational detection of decision-making capabilities is promising.

I. Introduction

In this paper, we consider a step toward a systematic approach to placing trust in a specific agent’s decision-making during operation of heterogeneous Cyber-Physical-Human (CPH) teams, with autonomous machine agents.

Resolving conflicts in a multiagent system’s operation can take various forms and depends on the application. Competition for resources (e.g., [1]) is one example, where multiple solutions are viewed in the context of Nash equilibrium. Maudet [2] gives an informative overview of the types of agreements that may be reached in multiagent systems. Theoretically, multiagent decision-making can be viewed as a sub-area of the vast subject of multiobjective or multicriteria decision-making and optimization, including game theory (see, e.g., [3]).

In future construction and exploration environments, such as lunar and planetary surfaces and undergrounds, transit to destinations in space, and persistent flight in planetary orbits and atmospheres, operations will have to rely on teams of many observers, controllers, and physical agents, both machine and human, both remote and present physically, working in concert toward a set of mission objectives. In complex environments with many decision makers, regardless of the deliberation process, a single decision must precede every action. At any timestep, which agent should have the authority for making that decision? On what grounds should authority assignment vary during operations? In other words, which agent should be trusted with the final decision that is likely to lead to a particular action and when?

We also seek answers to a related question: is it possible to precondition the system in terms of dynamic structure or the choice of decision makers at each point in time, so that the likelihood of positive outcomes of decision-making is maximized throughout the operation of the system? In a straightforward example, imagine that in sparse airspace, autonomous air vehicles proceed along optimal (great circle) trajectories to their destinations. In such sparse environments, the need for conflict-avoiding interactions is rare. In dense airspace with many vehicles converging on a single destination, organized traffic (e.g., along traffic streams) would be required. If the participating vehicles are equipped with sufficiently advanced sensors, decision-making algorithms, and physical capacity, such as high maneuverability, individually optimal traffic can be allowed to persist for longer durations than in the case of vehicles possessed of weaker situational awareness, slower computing, and lower agility. Thus, both the properties of the decision makers, machine or human, and the complexity of the environment can be adjusted to prevent reductions in efficiency and safety of the system before the onset of significant reductions. Similar considerations apply in any decision-making domain with multi-agent machine or human-machine teams.

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[†] Due to current policy of restricting method representation in publications, methods in this paper are described via text narrative rather than the customary pseudocode form.

Choosing the agent most likely to solve a given decision-making problem satisfactorily or reconfiguring the system itself to increase the chances of satisfactory outcome is a foundation of justifiably trusted systems. Definitions of trust abound (see, e.g., [3]). For the purposes of this paper, one agent trusts another when it accepts a decision and subsequent action directed by another agent, given some uncertainty in the outcome. Trust necessarily involves the notions of uncertainty and risk. If a positive outcome were guaranteed, no trust would be required.

In traditional, technical human-machine systems, the decision about the trusted agent is often made *a priori*. For instance, despite modern aircraft's ability to perform many functions without human assistance, the human pilot bears the ultimate authority and responsibility for the operation of aircraft. The exception to this rule, although not universal for all airlines, is also defined *a priori*: some of the automatic collision avoidance tools have priority over the pilot's decision-making.

While the relatively structured flight operations in our planet's airspace have so far been amenable to pre-defined rules of trust and authority, we conjecture that the new and harsh environments of space exploration, with their many unknowns, unpredictable equipment and software failures, and physiological and cognitive stress on human explorers, will require continual assessment of the trustworthiness of both human and machine CPH team members. Moreover, in reaction to changes in the quality of decision-making, the trust status and the attendant authority for particular instances of decision-making may have to be updated in the course of operations.

One aspect of trusting an agent with solving a problem is related to the history of that agent's performance in an environment of specific complexity. If the environment is expected to exceed a given complexity threshold, either the environment complexity must be changed or an agent with increased tolerance for complexity must be assigned authority for decision-making. Our approach to this aspect of the problem will rely on previous work [4].

Aside from informing the allocation of trust and, hence, authority in multiagent machine teams, we hope that the proposed approach will inform solutions to a related, timely problem – that of human-machine functional allocation. The latter has been an active and open-ended area of research, with frequently heuristic, context-dependent solutions. The general sentiment is usually expressed as “let machines do what they do best and humans what they do best”. Some activities possess a clear demarcation between the areas of human and machine abilities, for instance, large-scale physical models that have no known closed-form solutions are amenable only to machine computation. Other decision-making in complex, changing environments is not easily partitioned between humans and machines. We hope that formalization of trust and authority for decisions in *general* multiagent teams will offer criteria for dynamic functional allocation between humans and machines.

The following are the overarching working hypotheses of this investigation:

- *It is feasible to build actionable, computational models of problem-solving complexity for individual team members (human or machine) as a function of the system complexity parameters.* Traditionally, such models are assumed *a priori* and are static. For instance, a human air traffic controller is entrusted with handling no more than a predetermined maximum number of aircraft, based on the general estimate of a human's cognitive capacity. Ensuring that a system is controllable and safe is always based on bounding complexity. This will hold for systems with machine decision-makers. We are interested in assigning decision-making authority based on dynamic assessment of decision makers' reactions to arising complexity.
- *During operations, it is feasible to use information about the system and complexity models to predict approaches to phase transitions from safe and efficient operations to inefficient but safe operations to unsafe operations.* For instance, as system complexity increases, the decision-making problem becomes larger, i.e., incorporates more variables and constraints and possibly changing objectives and changing the fidelity of the physical models that govern the system's behavior, solutions of progressively larger, more computationally intensive problems will require more time and result in solutions of lower quality if limited to a time budget. We conjecture that the onset of complexity will be gradual and detectable.
- *It is possible to reconfigure the system to forestall phase transitions to undesirable states before they take place, by assigning decision-making authority to the agents best equipped to handle the changes in the environment or changing the system control structure.*

In the remainder of the paper, we state our assumptions, summarize the approach to reconfigurable system operation in response to dynamic complexities, and report on preliminary computational results on complexity trends for a simplified problem.

II. Problem Setting

In this initial work, we consider decision-making formalized as (generally nonlinear) optimization problems. Acknowledging the difficulty of posing many realistic decision-making problems as optimization formulations, we nonetheless believe that reasoning about decision-making requires a formalization of the problem and we conjecture that a formal representation of human-machine decision-making, albeit potentially incomplete, will serve to reduce uncertainty in the outcomes.

Preconditioning the system for better chances of successful decision-making rigorously poses many difficult problems in modeling and uncertainty quantification and propagation. Here are two salient ones:

- Implicit vs. explicit modeling. If we know or conjecture uncertainty bounds around an input variable, we can use, say, interval arithmetic to propagate that quantification of the input uncertainty to uncertainty bounds on the output of a computational model. Having no explicit access to human decision-making “arithmetic”, we must treat human decision-making “software” as a black box, with outputs of potentially high variability, based not just on differences among individual decision makers, but also on the variability of a single decision maker as a function of a multitude of internal and external variables, such as training and getting enough sleep before making decisions. Reliability of human decision-making is a vast area of research (for an excellent perspective, please see [5]). That said, research shows that human errors, and therefore reliability, are not random and take predictable and recurrent form [6]. At this stage, we are not modeling human error explicitly. Rather, we consider human error to be part of the overall performance. However, the finding that human errors are not random suggests that their occurrence can be reasonably incorporated into general problem-solving performance. Isolating the human error component of problem solving explicitly will be considered in subsequent works.
- Incompleteness of computational problem formulation. Because human decision-making usually involves both clearly stated and unstated (possibly subconscious) variables, objectives, and constraints, computational problem formulation may not—likely, does not—capture the same problem that the human decision maker is solving. Conversely, the human decision-maker is not likely to account for all problem components that equip computational team members. This may make the human vs. machine comparison of solutions difficult and subjective, unless overarching criteria of solution quality are accepted.

Recognizing these and many other complexities, in the initial proof of concept outlined in this paper, we focus on a simplified problem of *simulated* decision makers represented by optimization algorithms. As usual, turning to a simpler problem serves as a filter: should the approach prove infeasible on a simpler problem, it is not likely to be feasible on a more realistic one. In evaluating our simulated decision makers, we present simulated them with problems that are known to have solutions and consider the following behavior satisfactory:

1. A solution is found (time to solution is recorded) and solution criteria are satisfied.
2. A solution is not found, but the algorithm provides a justification for the result.

We consider the second criterion to be an idealized surrogate for “self-awareness” of a decision maker.

III. The Approach

In [4], we proposed a framework for complexity management in application to *ab initio* design of airspace traffic control. Here we state the approach applied to any time-dependent decision-making performed by a team of agents. For simplicity of exposition, here we consider decisions that individual agents make by solving a problem, without delving into the details of potential incorporation of “advice” from other agents. The case of integrating advice from other decision makers to inform an agent’s decision is relegated to other papers.

The basis of the approach is to consider a computable complexity measure as a combination of problem solvability on a time budget and the quality of solutions. Given that it is, in general, impossible to predict the amount of computational work required to solve a nonlinear optimization problem, we are not considering traditional measures of computational complexity, such as operation counts. Instead, we consider experimental performance of solvers (human or machine) and conjecture that for problems of specific structure, the growth of complexity in terms of quality and tractability of solutions on a time budget will follow predictable trends.

The concept of tractability on a time budget is straightforward and depends on the time-criticality and safety-criticality of the operational context. The concept of solution quality is also context-dependent but is more involved. The following are examples of general solution quality measures:

1. Constraint satisfaction. The ability of the decision maker to satisfy constraints to the desired convergence criterion. For constraints related to physical realizability of a process or safety constraints, satisfaction is a prerequisite.
2. Distance from optimality. Safety-critical and robustness factors being equal, a solution that is closer to optimality will be considered better than other solutions. Considering the example of moving along a trajectory, those maximally close to great circles are best, given constraint satisfaction.
3. Robustness. This is a context-dependent property. For a multi-agent system moving in space (2D or 3D), robust solutions could be those that are not likely to cause secondary or tertiary conflicts. In other words, a robust solution of this type maintains a maximally large feasible space for subsequent problems as it proceeds along its solution trajectory. Another expression of robustness is the maximum uncertainty under which performance is guaranteed.
4. Serendipity. Uncertainty about the environment of decision-making is not always harmful. For instance, taking a large exploration step in solving a decision-making problem may allow the solver to step over a local minimum to a better solution. Serendipity may be defined as the minimum uncertainty that would allow for great improvements in performance.

Specific realization of these and other measures of solution quality depends on the application domain. For instance, maintaining maximally large feasible space for subsequent positions would be a measure of solution robustness for vehicles moving in free flight.

The overall conceptual reasoning framework comprises two steps:

- Build complexity models.
 - Identify internal and external complexity parameters of the system that affect tractability and solution quality of the decision-making problem. In general, complexity parameters will have to do with problem size, smoothness, and nonlinearity, among others.

For instance, in the case of an aircraft's movement from point A to point B, some of the external system parameters include density and heterogeneity of traffic in a given volume of airspace, the available information about the topography and the motion of other aircraft. Some of the internal parameters are the physical properties of the aircraft, such as speed and maneuverability, the quality of sensors and communications, and the quality and speed of trajectory computation software.

In the example of multidisciplinary design optimization, system complexity parameters may include the number and fidelity of contributing disciplines, the multidisciplinary problem formulation, the quality of solvers, the number of design and state variables, and many others.

- For each agent in the system, use a context-dependent design of experiments to generate timing and solution quality data.

- Build a model of problem tractability and solution quality as a function of complexity variables, using an appropriate approximator, such as machine learning (ML). ML will be especially useful in building complexity models of human decision-makers, given the lack of access to explicit decision-making mechanisms in humans.

- Use complexity models during system operations to detect and forestall phase transitions to undesirable states.
 - Following system initialization, define the time intervals for assessing the system complexity and the temporal horizon along which system complexity is to be predicted.
 - At each time interval, acquire the current and predicted system parameters that inform the formulation of the decision-making optimization problem, translating to such properties as dimensionality, smoothness, and nonlinearity, among others.

- Input the system complexity parameters into complexity models of each decision-making agent to assess the ability to reach satisfactory solutions in terms of tractability and solution quality along the temporal prediction horizon.
- Given the predicted assessment of problem-solving potential, assess the need for system reconfiguration and reconfigure the system control, if needed. Reconfiguration can accommodate both increases and decreases in predicted complexity. The former may involve a reassignment of decision makers or restructuring of the system control, likely involving fewer degrees of freedom for individual decision makers. In contrast, the response to lowering complexity likely involves relaxation of constraints imposed on individual agents' decision-making.

Reconfiguration may be accomplished in several ways. Different decision makers may be employed to handle growing or diminishing complexity. For instance, if complexity involves the need for employing governing models of higher fidelity, variable-fidelity model management optimization algorithms may be engaged to solve the problem. If decreased complexity allows for a choice among a finite set of discrete decisions, a zero-order optimization solver may be chosen to select a solution. When the choice is between human and machine decision makers in the problem of authority allocation, complexity parameters may be characterized by dimensionality (e.g., the ability to handle a certain number of variables and objectives), the need for solving physics-based models quickly, or the ability to accurately assess visual information, among other parameters.

Another potential reconfiguration strategy deals with changing the system control, which means changing the problems decision makers must solve and potentially the locus of system control. Reconfiguration may be directed by an external authority, as may happen when “selfish” decisions (those optimal for individual decision makers) must be replaced with decisions that balance selfish optimality with system objectives. A more interesting option now under investigation is a change of the decision-making (and therefore optimization) problem formulation at the level of individual agents, based on information available to those agents, leading to an emergent improvement in the overall system performance.

Again, decisions on reconfigurations depend strongly on the operational context. For instance, operations with low time criticality allow for deliberation and negotiation among solutions. Stable problem complexity allows for *a priori*, rarely changing decision-making authority selection. High time criticality and safety criticality require *a priori* trust or assurance of reliable computations in real time. In all cases, formal representation of problem solving is needed in complex domains, to facilitate or even enable reasoning about decision-making in dynamic environments.

We note that, once a particular agent is equipped with a new capability (e.g., a faster, more robust optimization algorithm), the complexity model must be recalibrated.

IV. Preliminary Results: Performance of Simulated Decision Makers

In [4], we considered a simplified example of problem-solving complexity in *ab initio* operation of large numbers of aircraft in an airspace, without assuming any specific control strategy *a priori*. Here, we emulate general decision makers by three widely available, off-the-shelf optimization codes, to test whether trends in nonlinear programming problem tractability on a time budget and solution quality are detectable. In this case, all agents emulate machine decision-makers.

Because comparative evaluation of optimization software is not the purpose of this paper, we will name the optimization methods rather than the software. The simulated decision-makers are as follows:

- Opt1 decision maker represented by a feasible direction optimization code (e.g., [8]).
- Opt2 decision maker is represented by a Kreisselmeier–Steinhaus code (a penalty function approach, e.g., [9]).
- Opt3 decision maker is represented by a sequential least squares approach (a member of the sequential quadratic programming family, e.g., [10]).

These codes use algorithmically different approaches to solve constrained nonlinear optimization problems. Moreover, the codes provide different diagnostic information about the outcomes of optimization.

The data on tractability and solution quality is collected by solving 395 unconstrained and constrained optimization problems from the Hock and Schittkowski optimization test problem set [7]. Figure 1 shows the distribution of problems by the number of variables and constraints (here we denote the sum of variables and constraints by the term “combined dimension”).

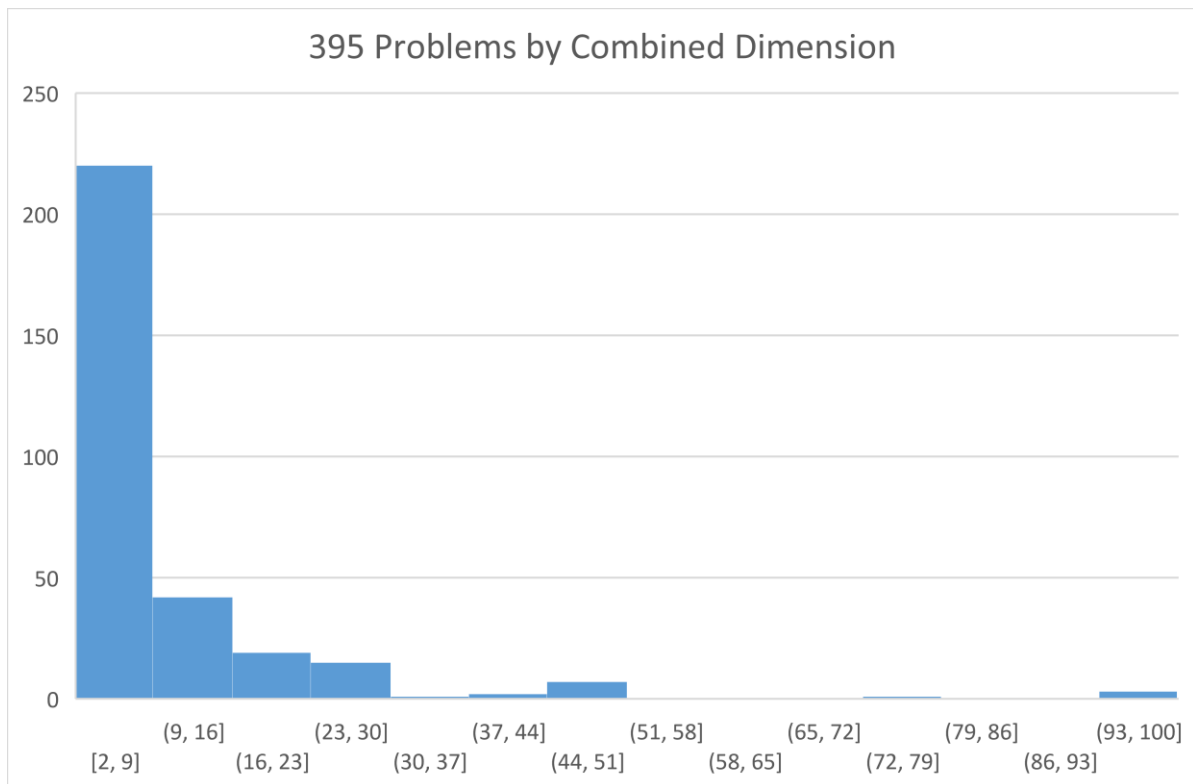


Figure 1 The number of problems by sum of variables and constraints.

Figures 2 and 3 summarize the temporal performance of the simulated decision makers for the entire set of problems and for the first 40 problems, respectively.

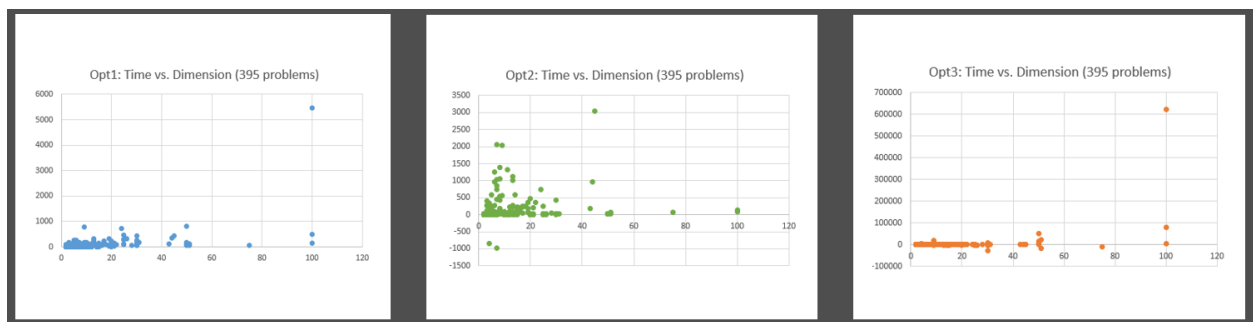


Figure 2 Time vs. Dimensions for the entire set of problems.

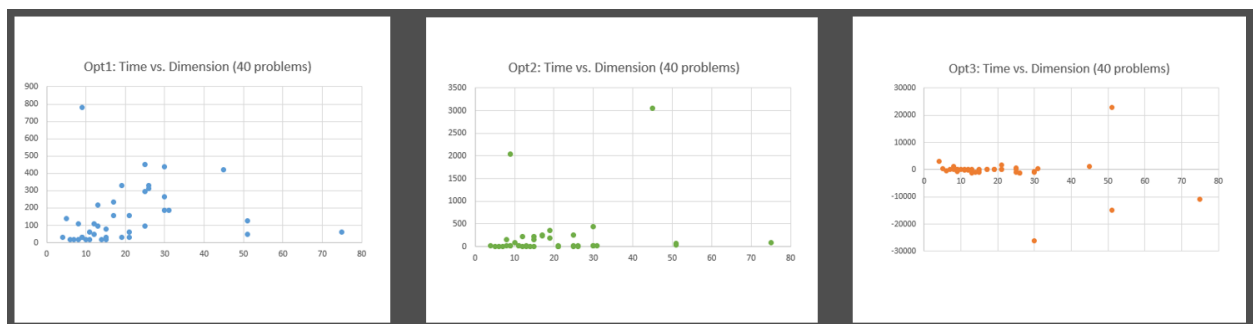


Figure 3 Time vs. Dimension for the first 40 problems.

The following notes summarize salient aspects of the performance:

- Opt1: The code declared success for all 395 problems; however, it completed with violated constraints for 77 problems.
- Opt2: The code declared success for 393 problems. It declared failure twice, due to exceeding the maximum allowed number of iterations, and completed computations with violated constraints for 39 problems.
- Opt3: The code declared success for 363 problems, declared failure due to violated constraints for 32 problems, and failure due to exceeding the maximum allowable number of iterations twice.

While detailed Time vs. Dimension analysis is work in progress, there are general trends in the increase of time to solution with larger problems. In addition, the nature of completion is indicative of some important aspects of decision-making behaviors that can inform the choice of agents to have the authority for making decisions in specific contexts. For instance, if self-awareness and accuracy have the highest priority, then Opt3 should be given authority, as it makes a distinction between a successful solution and one that violates the problem constraints. If never failing, but always providing the best available answer is the priority, then Opt1 is the winner, even if its solutions may have an unacceptable degree of uncertainty about realizability of the solution, given the failures of constraint satisfaction.

V. Concluding Remarks

This paper described initial steps toward addressing a critical aspect of uncertainty in the operation of complex, dynamic systems: the ability of participating agents to arrive at satisfactory decisions in time, as a function of problem complexity. This approach complements traditional uncertainty quantification and propagation by offering a way to precondition the decision-making system via detection of approaches to reduced solvability and reconfiguring decision-making or system control structure to forestall the onset of undesirable events.

Experiments with decision makers simulated by various optimization codes suggest that trends in tractability and quality of solutions as a function of problem complexity expressed by problem dimension and nonlinearities may be captured by data-fitting models.

The steps in progress are a detailed analysis of problem structure that affects tractability on a time budget; data collection on a use case of decision-making for a single problem structure where complexity is a function of size alone, followed by modifications in the problem objective set. We are also investigating meaningful use cases for inclusion and modeling the problem-solving capabilities of human decision makers.

Overall, the initial study of decision-making trends makes us cautiously optimistic that at least on the side of machine decision-making, practical management of complexity is feasible.

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